<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td>01.</td><td>Which of the following statements is defines a Server?
A. a device that only stores data
B. a system that provides services to clients on a network
C. a software used for virtualization
D. a network protocol</td><td>B. a system that provides services to clients on a network</td></tr>
<tr><td>02.</td><td>Which software component allows multiple operating systems to run on a single physical server from the following?
A. Network Operating System
B. Client OS
C. Hypervisor
D. RAID</td><td>C. Hypervisor</td></tr>
<tr><td>03.</td><td>According to the following RAID levels which one primarily used for data redundancy through mirroring?
A. RAID 1
B. RAID 0
C. RAID 5
D. RAID 10</td><td>A. RAID 1</td></tr>
<tr><td>04.</td><td>Which DNS record is used to map a domain name to an IPv4 address from the following?
A. MX
B. AAAA
C. CNAME
D. A</td><td>D. A</td></tr>
<tr><td>05.</td><td>Match the following DNS records with their functions
<table>
<tr><th>Column A</th><th>Column B</th></tr>
<tr><td>1. MX</td><td>A. Identifies the authoritative DNS server</td></tr>
<tr><td>2. CNAME</td><td>B. Maps a hostname to another hostname</td></tr>
<tr><td>3. NS</td><td>C. Specifies mail server</td></tr>
<tr><td>4. A</td><td>D. Maps hostname to IPv4 address</td></tr>
</table></td><td>1–C
2–B
3–A
4–D</td></tr>
</table>

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>06.</td><td>Match the following terms from column A with their descriptions in column B
<table>
<tr><th>Column A</th><th>Column B</th></tr>
<tr><td>1. Hypervisor</td><td>A. Manages and allocates IP addresses</td></tr>
<tr><td>2. DHCP</td><td>B. Provides directory services</td></tr>
<tr><td>3. Active Directory</td><td>C. Creates and manages virtual machines</td></tr>
<tr><td>4. DNS</td><td>D. Resolves domain names to IP addresses</td></tr>
</table></td><td>1–C

2–A

3–B

4–D</td></tr>
<tr><td>07.</td><td>A Network Operating System (NOS) is used to manage network resources and services (True or False).</td><td>True</td></tr>
<tr><td>08.</td><td>The nslookup command is used to check DHCP lease time on a client (True or False).</td><td>False</td></tr>
<tr><td>09.</td><td>What is the primary role of a server in a network?

A. To request services from clients
B. To provide resources and services to clients
C. To manage personal files
D. To replace a network operating system</td><td>B. To provide resources</td></tr>
<tr><td>10.</td><td>Which term refers to software that allows multiple operating systems to run on a single physical machine from the following?

A. Network Operating System
B. Server
C. Hypervisor
D. Client</td><td>C. Hypervisor</td></tr>
</table>

	QUESTIONS	Answers
11.	Which of the following is a key benefit of server virtualization? A. Increased hardware cost B. Reduced server utilization C. Improved resource utilization and flexibility D. Increased power consumption	C. Improved resource utilization and flexibility
12.	Which RAID level provides data redundancy by mirroring data across disks from the following? A. RAID 0 B. RAID 1 C. RAID 5 D. RAID 10	B. RAID 1
13.	According to the following functions, which one is the main function of Active Directory Domain Services (ADDS)? A. Assign IP addresses automatically B. Resolve domain names to IP addresses C. Manage users, computers, and security in a domain D. Monitor server performance	C. Manage users, computers, and security in a domain
14.	Which DNS record type is used to map a domain name to an IPv4 address from the following? A. AAAA B. MX C. PTR D. A	D. A
15.	From the following components which DHCP component defines the range of IP addresses that can be assigned to clients?	B. Scope

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td></td><td>A. Reservation
B. Scope
C. Failover
D. Lease time</td><td></td></tr>
<tr><td>16.</td><td>From the following command tools,Which command-line tool is commonly used to test and troubleshoot DNS name resolution?

A. ping
B. ipconfig
C. nslookup
D. tracert</td><td>C. nslookup</td></tr>
<tr><td>17.</td><td>Server virtualization helps organizations reduce hardware costs by allowing multiple virtual machines to run on a single physical server (True or False).</td><td>True</td></tr>
<tr><td>18.</td><td>A hypervisor works above the guest operating systems and controls how hardware resources are shared among virtual machines (True or False).</td><td>True</td></tr>
<tr><td>19.</td><td>DNS is responsible for assigning IP addresses dynamically to client computers on a network (True or False)..</td><td>False</td></tr>
<tr><td>20.</td><td>Match the server components from column A with their purposes in column B

<table><tr><th>Column A</th><th>Column B</th></tr><tr><td>1. Hypervisor</td><td>A. Converts domain names into IP addresses</td></tr><tr><td>2. DHCP</td><td>B. Hosts and manages user authentication and policies</td></tr><tr><td>3. DNS</td><td>C. Allocates IP addresses automatically</td></tr><tr><td>4. Domain Controller</td><td>D. Manages virtual machines on a server</td></tr></table></td><td>1–D

2–C

3–A

4–B</td></tr>
</table>

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td>21.</td><td>Match the RAID levels from column A with their characteristics in column B

<table><tr><th>Column A</th><th>Column B</th></tr><tr><td>1. RAID 0</td><td>A. Improves fault tolerance using parity</td></tr><tr><td>2. RAID 1</td><td>B. No redundancy, improved performance</td></tr><tr><td>3. RAID 5</td><td>C. Data mirroring for redundancy</td></tr><tr><td>4. RAID 10</td><td>D. Combines mirroring and striping</td></tr></table></td><td>1–B

2–C

3–A

4–D</td></tr>
<tr><td>22.</td><td>From the following roles Why is a Network Operating System (NOS) important in server administration?

A. It allows users to create personal documents
B. It manages network resources and enables communication between clients and servers
C. It replaces the need for hardware components
D. It is used only for installing applications</td><td>B. It manages network resources and enables communication between clients and servers</td></tr>
<tr><td>23.</td><td>According to the following statements How does a hypervisor support server virtualization?

A. By acting as a firewall between networks
B. By managing user accounts and permissions
C. By allowing multiple virtual machines to share the same physical hardware
D. By converting IP addresses into domain names</td><td>C. By allowing multiple virtual machines to share the same physical hardware</td></tr>
<tr><td>24.</td><td>According to the following Technology, why is RAID technology used in server storage systems?

A. To increase internet speed
B. To combine multiple disks for improved performance, reliability, or fault tolerance
C. To reduce the need for backups
D. To eliminate the need for operating systems</td><td>B. To combine multiple disks for improved performance, reliability, or fault tolerance</td></tr>
</table>

	QUESTIONS	Answers
25.	From the following network environment, how does Active Directory Domain Services (ADDS) help in managing a network environment? A. By assigning IP addresses to devices B. By providing web hosting services C. By centralizing authentication, authorization, and management of network resources D. By monitoring network traffic only	C. By centralizing authentication, authorization, and management of network resources
26.	From the following Reasons why is a DHCP scope configured on a server? A. To resolve domain names to IP addresses B. To define and control the range of IP addresses that can be leased to clients C. To store DNS zone files D. To promote a server to a domain controller	B. To define and control the range of IP addresses that can be leased to clients
27.	A company wants to run three different operating systems on one physical server. Which component should be installed first to achieve this from the following? A. Network Operating System B. Guest Operating System C. Hypervisor D. Active Directory Domain Services	C. Hypervisor
28.	An administrator wants to ensure data availability if one hard disk fails. From the following RAID, which RAID level should be configured on the physical server? A. RAID 0 B. RAID 1 C. RAID 0+0 D. RAID 6	B. RAID 1
29.	A client computer cannot access websites using domain names but can access them using IP addresses. From the following roles which server role should the administrator	C. DNS

	QUESTIONS	Answers
	check and configure? A. DHCP B. Active Directory C. DNS D. Hypervisor	
30.	An administrator needs to assign the same IP address to a specific client every time it connects to the network. Which of the following DHCP feature should be used? A. Reservation B. Exclusion C. Scope D. Lease time	A. Reservation
31.	After installing Active Directory Domain Services, which action is required to make the server manage users and computers in the following domain? A. Install DNS B. Create a virtual machine C. Configure RAID D. Promote the server to a domain controller	D. Promote the server to a domain controller
32.	A company wants to run multiple operating systems on one physical server to reduce hardware costs. Which of the following technologies should the administrator apply? A. Active Directory Domain Services B. Server virtualization using a hypervisor C. DNS zone configuration D. DHCP failover	B. Server virtualization using a hypervisor
33.	An administrator needs to protect critical server data against disk failure by duplicating data across two drives. Which of the following RAID level should be configured? A. RAID 0 B. RAID 1	B. RAID 1

	QUESTIONS	Answers
	C. RAID 5 D. RAID 6	
34.	After installing Windows Server, an administrator wants to centrally manage users and computers in the network. Which of the following action should be performed next? A. Install DHCP role B. Configure RAID storage C. Install ADDS and promote the server to a domain controller D. Configure DNS lookup zones	C. Install ADDS and promote the server to a domain controller
35.	A client computer is unable to resolve a domain name to an IP address. Which of the following command should the administrator use to diagnose the problem? A. ipconfig B. nslookup C. netstat D. tasklist	B. nslookup
36.	To ensure specific devices always receive the same IP address from the DHCP server, which of the following DHCP feature should be applied? A. Scope exclusion B. Lease time C. Reservation D. Failover	C. Reservation
37.	Which of the following type of user account has full control over system settings and resources? A. Standard B. Guest C. Administrator D. Temporary	C. Administrator

	QUESTIONS	Answers
38.	Which of the following actions is performed to temporarily prevent a user from logging into the system without deleting the account? A. Remove account B. Deactivate account C. Change password D. Copy account	B. Deactivate account
39.	What does OU stand for in the following Active Directory management? A. Operating Unit B. Organizational User C. Organization Unit D. Organizational Unit	D. Organizational Unit
40.	Which of the following tool is used to create and manage Group Policy Objects (GPOs)? A. Group Policy Management Console (GPMC) B. Control Panel C. Server Manager D. Task Manager	A. Group Policy Management Console (GPMC)
41.	Which the following types of user account has full control over the system and can manage other user accounts? A. Standard B. Guest C. Administrator D. Temporary	C. Administrator
42.	What is the term used for organizing users, computers, and other objects within Active Directory from the following? A. User Group B. Organization Unit (OU) C. Domain Controller D. Group Policy Object	B. Organization Unit (OU)

	QUESTIONS	Answers
43.	Which of the following tools is used to create and manage Group Policy Objects (GPOs)? A. Task Manager B. Server Manager C. Group Policy Management Console (GPMC) D. Control Panel	C. Group Policy Management Console (GPMC)
44.	Which of the following actions allows a client computer to become part of a domain? A. Changing the user password B. Creating a new user account C. Joining the client computer to the domain D. Assigning permissions to a user	C. Joining the client computer to the domain
45.	Copying a user account helps create a new account with similar permissions and settings as an existing account **(True or False).**.	True
46.	Organizational Units (OUs) are used to apply Group Policy settings only at the domain level **(True or False).**	False
47.	Joining a client computer to a domain requires proper connectivity between the client and the server **(True or False).**	True
48.	According to the following polices, why are user account policies important in server administration? A. They are used only to create new user accounts B. They help control security by defining password rules and access behaviour C. They allow users to install server roles D. They replace group policies	B. They help control security by defining password rules and access behaviour
49.	How does placing users into groups help administrators manage user accounts from the following?	C. It simplifies permission management by assigning rights to groups instead of individual users

	QUESTIONS	Answers
	A. It increases the number of user accounts B. It allows administrators to assign permissions individually to every user C. It simplifies permission management by assigning rights to groups instead of individual users D. It prevents users from logging in to the domain	
50.	From the following statements, why are Organizational Units (OUs) used in Active Directory? A. To install operating systems on client computers B. To store DNS records C. To organize users and computers for easier management and policy application D. To replace domains	C. To organize users and computers for easier management and policy application
51.	An administrator wants a new employee to have the same permissions as an existing staff member. Which of the following action should be performed? A. Create a standard account B. Copy the existing user account C. Deactivate the existing account D. Remove the existing account	B. Copy the existing user account
52.	A user should no longer access the network temporarily, but their account must remain for future use. From the following Tasks What should the administrator do? A. Delete the user account B. Change the user password C. Move the user to another OU D. Deactivate the user account	D. Deactivate the user account
53.	An administrator needs to apply specific security settings only to computers in a particular department. What is the best action from the following? A. Apply the policy at the domain level B. Create an Organizational Unit (OU) and link a GPO to it	B. Create an Organizational Unit (OU) and link a GPO to it

	QUESTIONS	Answers
	C. Add users to a new group only D. Change user account passwords	
54.	An administrator wants to give several users the same access permissions without configuring each account individually. What should the administrator do from the following? A. Create separate user accounts for each user B. Assign permissions directly to each user C. Create a group and add users to the group D. Place users in different Organizational Units	C. Create a group and add users to the group
55.	A user is unable to log in because their account is temporarily blocked. From the following Activities which administrative action should be applied to restore access without deleting the account? A. Remove the user account B. Change the user's group membership C. Deactivate the user account D. Activate the user account	D. Activate the user account
56.	An administrator needs to apply specific security settings only to users in one department. From the following approach which one should be used? A. Apply a Local Group Policy to all computers B. Create a GPO and link it to the department's Organizational Unit (OU) C. Change the domain password policy D. Assign administrator privileges to the users	B. Create a GPO and link it to the department's Organizational Unit (OU)
57.	From the following statements which one describe the meaning of a web server? A. A device used to design websites B. A system that stores and delivers web pages to users over the internet	B. A system that stores and delivers web pages to users over the internet

	QUESTIONS	Answers
	C. A software used only for databases D. A network security tool	
58.	Which of the following is a web server developed by Microsoft? A. Apache B. Nginx C. IIS D. Lighttpd	C. IIS
59.	Which feature must be enabled in IIS to support ASP.NET web applications from the following? A. FTP Feature B. CGI Interface C. ASP.NET D. HTTP Health and Diagnostics	C. ASP.NET
60.	What is the primary function of a web server from the following? A. To create user accounts B. To host and serve web content to clients over the internet or network C. To manage databases only D. To perform antivirus scanning	B. To host and serve web content to clients over the internet or network
61.	Which of the following is an example of a web server software? A. Microsoft Word B. Apache C. Photoshop D. Windows Explorer	B. Apache
62.	Which of the following IIS feature is used to enable execution of web applications built using ASP.NET? A. FTP Feature B. CGI Interface	C. Enable ASP.NET

	QUESTIONS	Answers
	C. Enable ASP.NET D. HTTP Health and Diagnostics	
63.	Why HTTP features must be enabled when configuring IIS on a Windows Server from the following? A. To store website files B. To allow the server to process and deliver web requests C. To configure DNS zones D. To manage user accounts	B. To allow the server to process and deliver web requests
64.	From the Reasons bellow, Why site binding is important when hosting a website using HTTP or HTTPS? A. It controls how users upload files to the server B. It disables unused web services C. It increases website speed automatically D. It links the website to a specific IP address, port number, and protocol	D. It links the website to a specific IP address, port number, and protocol
65.	From the following functions why is site binding important in IIS when configuring a web application? A. It increases the storage capacity of the server B. It maps a website to specific protocols, IP addresses, and port numbers for proper access C. It automatically creates user accounts D. It installs backend software	B. It maps a website to specific protocols, IP addresses, and port numbers for proper access
66.	From the following roles,how does choosing between free and paid hosting platforms affect a web application? A. It determines only the programming language of the application B. It affects the availability of server resources, technical support, and backend capabilities C. It changes the website's URL automatically D. It removes the need for a web server	B. It affects the availability of server resources, technical support, and backend capabilities
67.	An administrator wants to deploy a PHP web application on IIS. Which of the following steps should be performed? A. Configure FastCGI handler mapping and PHP environment variables	A. Configure FastCGI handler mapping and PHP environment variables

	QUESTIONS	Answers
	B. Enable ASP.NET only C. Install Node.js module only D. Set up FTP without enabling HTTP features	
68.	A web developer wants to make a site accessible via a specific domain name and IP address. Which of the follwonwing IIS configuration should be used from the following? A. Enable HTTP Health and Diagnostics B. Configure site binding with the IP address, protocol, and DNS C. Install CGI Interface D. Verify website size	B. Configure site binding with the IP address, protocol, and DNS
69.	You have developed a PHP web application locally and want to host it on IIS. Which of the following actions should you take to ensure it runs correctly? A. Enable HTTP Features, Configure PHP Environment variables, and set up FastCGI handler mapping B. Install Node.js only C. Change the web browser settings D. Disable HTTP Features and FTP	A. Enable HTTP Features, Configure PHP Environment variables, and set up FastCGI handler mapping
70.	A website needs to be accessible online using HTTPS. Which of the following IIS configuration should be applied? A. Configure site binding with HTTPS protocol, assign the IP address, and configure the port number B. Upload the web app without specifying protocols C. Only enable FTP D. Disable all security features	A website needs to be accessible online using HTTPS. Which IIS configuration should be applied?
71.	Explain the steps you would take to set up a fully functional Windows server environment for a small organization, including hardware and software requirements, installation of the server OS, creation of virtual machines, configuration of RAID, installation of	**1.** Cost Savings ✓ Reduces the need for physical hardware because multiple virtual

	QUESTIONS	Answers
	Hypervisor, setup of DNS and DHCP services, and promotion of the server to a domain controller. Discuss at least five benefits of server virtualization in your setup	machines (VMs) can run on a single physical server. ✓ Lowers energy consumption and physical space requirements. **(2 Mark)** **2.** Better Resource Utilization ✓ Virtualization allows servers to run at higher utilization rates. ✓ CPU, memory, and storage resources can be dynamically allocated to different VMs based on demand. **(2 Mark)** **3.** Simplified Management ✓ Centralized management of virtual machines through hypervisor platforms. ✓ Easier to monitor, maintain, and update servers without physical access. **(2 Mark)** **4.** Rapid Deployment and Scalability ✓ New virtual servers can be deployed quickly without buying additional hardware. ✓ Virtual environments can scale easily to meet business demands.

	QUESTIONS	Answers
		(2 Mark) **5.** Isolation and Security ✓ Each VM runs independently; a failure or compromise in one VM does not affect others. ✓ Helps in testing and development environments without risking production systems. **(2 Marks)** **6.** Disaster Recovery and Backup ✓ Virtual machines can be backed up as images, making recovery faster and simpler. ✓ Snapshots allow easy rollback to previous states in case of errors. **(2 Mark)** **7.** Flexibility and Compatibility ✓ Multiple operating systems can run on the same physical server. ✓ Enables legacy applications to run alongside modern applications without conflicts. **(2 Marks)**

	QUESTIONS	Answers
		8. Improved Business Continuity ✓ Supports live migration of VMs to other physical servers without downtime. ✓ Enhances uptime and reliability for critical services. **(2 Marks)** **(Consider five points)**
72.	A company is experiencing frequent server downtime and slow performance after deploying several applications on a single physical server. Analyse the situation and explain how implementing server virtualization with an appropriate hypervisor and RAID configuration could help resolve these issues. In your answer, consider performance, availability, and resource management.	• Analyse the problem and identify that downtime and poor performance are caused by multiple applications competing for resources on a single physical server. **(2Marks)** • Implement a hypervisor Install a hypervisor to enable server virtualization on the physical machine. **(2Marks)** • Create and isolate virtual machines Run each application in its own virtual machine to improve stability and availability. **(2Marks)** • Optimize resources and storage Allocate CPU, memory, and storage efficiently and configure RAID for data redundancy and fault tolerance. **(2Marks)** • Evaluate improvements Confirm improved performance, reduced downtime, better resource management, and

	QUESTIONS	Answers
		higher system reliability. **(2Marks)**
73.	A company notices that some users have more access than necessary for their roles, and certain critical settings are being modified without authorization. Analyse the scenario and determine which tools and configurations you would review to identify and correct these issues	Review user account levels (Standard vs. Administrator), **(3Marks)** assigned permissions, group memberships, Organizational Units (OUs), **(3Marks)** and Group Policy Objects (GPOs) to identify misconfigurations or excessive access rights. **(4Marks)**
74.	An administrator needs to organize users in a large organization and ensure that specific policies are applied only to certain departments. Analyse how you would use Organizational Units (OUs) and Group Policy Objects (GPOs) to achieve this	1.Create OUs representing departments (**3Marks)** 2.Assign users to respective OUs, and link appropriate GPOs to each OU to apply policies selectively. (**3Marks)** 3.Examine the hierarchy and inheritance of GPOs to avoid conflicts. **(4Marks)**
75.	A company has multiple departments with different access needs. Some users report they cannot access certain resources, while others have unnecessary access to sensitive files. Analyze the situation and explain how using Organizational Units (OUs), groups, and Group Policy Objects (GPOs) could help the administrator resolve these access issues. Include considerations for security, efficiency, and ease of management	**Step 1**: Identify the problem Some users cannot access needed resources, while others have unnecessary access, indicating mismanaged permissions and poor organization of accounts. **(2Marks)** **Step 2**: Organize users into Organizational Units (OUs) Create OUs for each department or functional group. This helps segment users logically based on their role or department. **(2Marks)**

	QUESTIONS	Answers
		Step 3: Create user groups within OUs Assign users to appropriate groups within their OU. Groups allow administrators to manage permissions collectively rather than individually. **(2Marks)** **Step 4**: Apply Group Policy Objects (GPOs) Create and link GPOs to each OU or group to enforce security settings, access rights, and restrictions specific to the department's needs. **(2Marks)** **Step 5**: Review and adjust permissions Grant or revoke access rights as needed, ensuring that users have only the permissions required for their tasks. This improves security, efficiency, and manageability. **(2Marks)**
76.	A company notices that some users have inappropriate access to resources, while others cannot access the resources they need. Analyse how the administrator could use user account management (activation, deactivation, password changes) and group management (creating groups, adding/removing users) to resolve these issues. Include considerations for security and efficiency	**Step 1**: Identify the problem • Review which users have inappropriate access or cannot access required resources. • Determine whether the issue is due to account status (active/deactivated) or group membership. **(2Marks)**

	QUESTIONS	Answers
		Step 2: Manage user accounts • Activate accounts for users who need access. • Deactivate accounts for users who should no longer have access. • Change passwords if accounts are compromised or require security updates. • Remove accounts if users have permanently left the organization. **(2Marks)** **Step 3:** Manage user groups • Create groups that correspond to different roles or access levels. • Add users to appropriate groups based on their responsibilities. • Remove users from groups where they should no longer have access. **(2Marks)** **Step 4:** Customize account parameters • Apply settings such as password expiration, login restrictions, and access permissions to enhance security and usability. **(2Marks)**

	QUESTIONS	Answers
		Step 5: Review and verify • Test user access to ensure the correct permissions are applied. • Regularly audit accounts and group memberships to maintain security and efficiency. **(2Marks)**
77.	Your organization is considering whether to grant remote access permissions to all administrators or only a few trusted accounts. Evaluate the risks and benefits of each approach and recommend the best course of action	1.Granting remote access to all administrator's increases convenience but raises security risks. **(4Marks)** 2.Limiting access reduces risk of unauthorized changes and potential breaches. **(3Marks)** 3. Recommend granting remote access only to trusted, trained administrators and monitoring their activities. **(4Marks)**
78.	An organization is reviewing its current user account and group management strategy. Evaluate whether using individual user permissions or managing access primarily through groups and Organizational Units (OUs) is more effective. Consider factors such as security, administrative efficiency, scalability, and ease of policy enforcement in your evaluation	• **Groups and OUs**: Simplify administration, allow consistent policy application via GPOs, scalable for large organizations, easier to manage new users. **(3 Marks)** • **Individual permissions**: Provide granular control, but can be time-consuming, prone to errors, and difficult to maintain as the organization grows. **(3 Marks)** • **Evaluation Conclusion**: Using groups and OUs is generally more

	QUESTIONS	Answers
		effective for balancing security, efficiency, and scalability, while individual permissions may only be suitable for very specific or temporary needs. **(4 Marks)**
79.	You are reviewing the group and user management strategy for your domain. Evaluate the effectiveness of using groups versus individual user permissions for managing access to shared resources	1.Using groups simplifies permission management, ensures consistency, and reduces errors. **(3Marks)** 2.Individual permissions offer more granular control but are harder to maintain and prone to mistakes. **(3Marks)** 3.Evaluate which method balances efficiency and security best for the organization **(4Marks)**
80.	A company is deciding whether to manage user permissions by assigning rights to individual accounts or by using groups and OUs. Evaluate the pros and cons of each approach and recommend the best method for a large organization. Justify your answer	Managing permissions via individual accounts vs. groups and OUs **Step 1**: Identify both approaches • Individual account permissions: Assign rights directly to each user. **(2Marks)** • Groups and OUs: Assign users to groups and OUs, then apply permissions collectively. **(2Marks)** **Step 2**: Evaluate pros and cons • Individual accounts:

	QUESTIONS	Answers
		❖ Pros: Fine-grained control for specific users. ❖ Cons: Time-consuming, error-prone, difficult to scale in large organizations. **(2Marks)** • Groups and OUs: ❖ Pros: Efficient management, easier to apply consistent policies, scalable, reduces errors. ❖ Cons: Requires careful planning of structure. **(2Marks)** **Step 3:** Recommendation Use groups and OUs because they provide better scalability, consistency, and efficiency for large organizations. **(2Marks)**
81.	After implementing Group Policy Objects (GPOs) across several departments, some users report conflicts with local settings on their computers. Evaluate the effectiveness of the current GPO implementation and suggest strategies to improve policy management and reduce conflicts.	**Step 1:** Identify the problem Users experience conflicts between GPOs and local settings, which can cause inconsistent behaviour or errors. **(2Marks)** **Step 2:** Evaluate effectiveness • Properly applied GPOs enforce organizational policies and security. • Conflicts reduce effectiveness, showing the need for review and

	QUESTIONS	Answers
		hierarchy management. **(3Marks)** **Step 3:** Suggest improvements • Review GPO hierarchy (local, site, domain, OU) to ensure correct precedence. • Use starter GPOs and testing environments before deployment. • Document GPO settings and monitor for conflicts. **(3Marks)** **Step 4:** Conclusion Careful planning and testing improve effectiveness and minimize conflicts. **(2Marks)**
82.	An administrator is considering deactivating unused user accounts versus deleting them permanently. Evaluate the potential risks and benefits of each approach and recommend which strategy would better support security and organizational management.	**Step 1: Identify both options** • Deactivate accounts: Temporarily disable access without removing data. • Delete accounts: Permanently remove user accounts and data. **(2Marks)** **Step 2:** Evaluate risks and benefits • Deactivation: ❖ Pros: Preserves historical data, can reactivate if needed, safer for audit compliance.

	QUESTIONS	Answers
		❖ Cons: Occupies system resources, may be overlooked. **(2Marks)** • Deletion: ❖ Pros: Frees resources, removes unnecessary accounts. ❖ Cons: Data loss, difficult to recover, may violate compliance policies. **(4Marks)** **Step 3:** Recommendation Use deactivation for unused accounts, especially in environments requiring audit trails or historical data. Delete only after confirming data is no longer needed. **(2Marks)**
83.	A web application deployed on IIS is not accessible from certain browsers, and users report slow response times. Analyse the possible causes using IIS configuration and server environment factors.	Examine site bindings (HTTP/HTTPS, IP addresses, port numbers), **(3Marks)** handler mappings, HTTP features, backend configuration (PHP/Node.js), **(3Marks)** network connectivity, server security settings, and verify the local and remote accessibility tests to identify the cause of the problem. **(4Marks)**
84.	You are tasked with choosing a web hosting platform for a new application. Analyse the technical requirements and differences between free hosting and paid hosting, with and without a backend server	Free vs Paid Web Hosting (With and Without Backend)

	QUESTIONS	Answers
		Choosing a web hosting platform depends mainly on cost, technical requirements, and whether the application uses a backend server. **(2Marks)** • Free hosting is best for static websites or learning projects. It offers no-cost deployment, simple setup, and basic features like HTTPS and CDN, but has limited resources, low scalability, restricted backend support, and no guaranteed uptime or support. **(2Marks)** • Paid hosting provides better performance, reliability, security, and scalability. It supports backend servers, databases, continuous uptime (SLAs), and professional support, making it suitable for production and business applications. **(2Marks)** • Applications without a backend (static frontend only) can run efficiently on free hosting platforms. **(2Marks)** • Applications with a backend server require paid hosting to ensure stable server processes, database persistence, security, and scalability. **(2Marks)**

	QUESTIONS	Answers
85.	A web application hosted on an IIS server is running slowly and some users cannot access certain pages. Analyse the possible causes related to server configuration, protocols, and web app environment, and suggest steps to identify and resolve the issues	**Step 1:** Identify potential issues • Check server performance: CPU, memory, disk usage. • Verify site binding and protocol configuration (HTTP/HTTPS, IP, port). • Inspect web app environment: PHP/Node.js configuration, FastCGI handler mapping, IIS modules. • Examine network connectivity and firewall settings. **(2 Marks)** **Step 2:** Analyze IIS configuration • Verify HTTP Features, ASP.NET, and CGI/handler mappings are properly enabled. • Check FTP or file permissions if resources are not loading. **(2 Marks)** **Step 3:** Analyze web app environment • Ensure backend technologies (PHP, Node.js) are configured correctly. • Check for missing environment variables or misconfigured

	QUESTIONS	Answers
		caches (WinCache, IISNode). **(2 Marks)** **Step 4: Suggest corrective actions** • Optimize server resources and restart services if needed. • Correct site bindings, protocols, and ports. • Update handler mappings and environment configurations. • Test accessibility and performance after changes. **(4 Marks)**
86.	An organization is deciding whether to host its web application on a free hosting platform or a paid Windows Server with IIS. Analyze the technical and security implications of both options and explain which would be more suitable for a business-critical application.	**Step 1: Identify requirements** • Determine business-critical needs: security, reliability, performance, backend support, and scalability. **(2Marks)** **Step 2: Analyze free hosting** • Pros: Low/no cost, quick setup. • Cons: Limited resources, limited backend support, shared environment, lower security and control. **(2Marks)** **Step 3:** Analyze paid Windows Server with IIS

	QUESTIONS	Answers
		• Pros: Full control over server configuration, supports backend technologies (PHP, Node.js), enhanced security, scalable, custom DNS and port configuration. • Cons: Higher cost, requires server administration skills. **(2Marks)** **Step 4: Compare options** • Free hosting may suffice for simple or testing applications. • Paid IIS hosting is better for production, business-critical applications that require security, performance, and flexibility**. (2Marks)** **Step 5: Recommendation** • Use paid Windows Server with IIS for business-critical web applications to ensure reliability, security, and proper environment configuration. **(2 Marks)**
87.	An organization is deciding whether to assign Administrator accounts to department heads or to use Standard user accounts with delegated control and Group Policy Objects (GPOs)**.** Evaluate this decision by comparing the security,	Administrator vs Standard Accounts with Delegation & GPO **Step 1:** Identify options

	QUESTIONS	Answers
	manageability, and risk implications of both approaches, and justify which option is more appropriate for a domain-based environment	• Administrator accounts vs Standard accounts with delegation **(2 Marks)** **Step 2:** Evaluate security • Administrator → high risk • Standard + delegation → controlled access **(2 Marks)** **Step 3:** Evaluate management • Admin accounts are hard to manage • GPOs enable centralized control **(2Marks)** **Step 4:** Evaluate risk • Full admin rights increase system vulnerability **(2 Marks)** **Step 5:** Final judgment Standard accounts with delegation and GPOs are better**(2 Marks)**
88.	A network administrator is managing users using individual user permissions**,** user groups, and Organizational Units (OUs) with linked Group Policy Objects (GPOs)**.** Critically evaluate the effectiveness of managing permissions through groups and OUs with GPOs versus	Individual Permissions Vs Groups, OUs & GPOs **Step 1:** Identify approaches

	QUESTIONS	Answers
	assigning permissions directly to individual user accounts, and justify which method provides better scalability and control in a large organization.	• Individual user permissions vs Group/OU-based management **(2 Marks)** **Step 2:** Evaluate scalability • Individual permissions do not scale well • Groups and OUs handle many users efficiently **(2 Marks)** **Step 3:** Evaluate consistency • Individual setup causes errors GPOs enforce uniform policies **(2 Marks)** **Step 4:** Evaluate administration Group-based management saves time **(2 Marks)** **Step 5:** Final judgment Groups, OUs, and GPOs are more effective**(2 Marks)**

	QUESTIONS	Answers
89.	A company is considering hosting its web application on IIS versus Apache or Nginx. Evaluate the benefits and drawbacks of each web server in terms of performance, security, ease of configuration, and compatibility with backend technologies. Which option would you recommend for a high-traffic business application, and why?	Evaluating IIS vs Apache vs Nginx for hosting a web application **Step 1:** Identify criteria for evaluation • Performance (handling concurrent requests, speed) • Security (built-in protections, updates) • Ease of configuration and management • Compatibility with backend technologies (PHP, Node.js, ASP.NET) **(2 Marks)** **Step 2:** Evaluate IIS • Pros: Excellent integration with Windows Server, supports ASP.NET and other Microsoft technologies, easy GUI management. • Cons: Licensing cost, less common on Linux, heavier resource usage. **(2 Marks)** **Step 3:** Evaluate Apache

	QUESTIONS	Answers
		• Pros: Cross-platform, widely used, strong community support, supports PHP and other backends. Cons: Slightly slower than Nginx for high concurrency, more complex configuration for advanced features. **(2 Marks)** **Step 4: Evaluate Nginx** • Pros: High performance and low resource usage, excellent for handling many simultaneous connections, supports reverse proxy and caching. • Cons: Steeper learning curve for configuration, less GUI-based management. **(2 Marks)** **Step 5: Recommendation** • For a high-traffic business application on Windows infrastructure, **IIS** is recommended if using ASP.NET and Windows Server. • For Linux-based high-concurrency applications, **Nginx** is preferable. • Apache is suitable for general-purpose hosting with moderate traffic.**(2 Marks)**

	QUESTIONS	Answers
90.	After deploying a web application on a Windows Server using IIS, the company notices occasional downtime and slow website speed. Evaluate whether the issues are more likely caused by server configuration, web app design, or network limitations. Recommend strategies to improve reliability and performance.	Evaluating causes of downtime and slow website speed on IIS **Step 1:** Identify possible causes • Server configuration**:** Improper IIS setup, incorrect site bindings, misconfigured HTTP/HTTPS protocols, insufficient resources. **(2 Marks)** • Web app design: Inefficient code, heavy scripts, unoptimized database queries. • Network limitations**:** Bandwidth constraints, firewall issues, DNS misconfiguration **(2 Marks)** **Step 2**: Evaluate likelihood of each cause • If downtime occurs under high load, likely server resource limitations or web app design issues**.** • If users cannot reach the site intermittently, likely network or DNS issues**. (2 Marks)** **Step 3:** Recommend strategies

	QUESTIONS	Answers
		Optimize IIS configuration**:** Adjust site bindings, enable HTTP/HTTPS properly, configure caching (WinCache, IISNode). Optimize web app: Minimize scripts, optimize database calls, implement caching. Check network: Ensure sufficient bandwidth, verify firewall rules, configure DNS properly. **(2 Marks)** **Step 4:** Monitor and test Continuously monitor server and network performance, perform load testing, and check accessibility and website speed. **(2 Marks)** **Step 5:** Conclusion Combining server, application, and network optimization will improve reliability and performance of the web application. **(2 Marks)**
91.	Design a strategy for managing user accounts, groups, and organizational units (OUs) in a medium-sized organization. Your strategy should ensure efficient permission management, security, and ease of administration. Explain the steps you would take and the tools you would use.	Designing a strategy for managing user accounts, groups, and OUs **Step 1:** Analyze the organization Identify the departments, roles, and access requirements of users. Determine which resources each group needs to access. **(3 Marks)**

	QUESTIONS	Answers
		Step 2: Create Organizational Units (OUs) Organize users into OUs based on departments, teams, or functional roles. This allows policies and permissions to be applied systematically. **(3 Marks)** **Step 3:** Create groups within OUs For each OU, create groups such as "Read-Only," "Standard User," and "Administrators." Assign users to these groups according to their role and required access. **(3 Marks)** **Step 4:** Assign permissions to groups Grant or revoke access rights at the group level rather than individual accounts. This ensures consistent permissions and simplifies management. **(3 Marks)** **Step 5:** Implement administrative policies and monitoring Use Group Policy Objects (GPOs) to enforce password policies, account restrictions, and security settings. Regularly audit accounts, groups, and

	QUESTIONS	Answers
		permissions to maintain security and efficiency. **(3 Marks)**
92.	Create a Group Policy Object (GPO) plan to enforce security settings across multiple departments in a company. Include how you would structure the GPOs, assign them to OUs or groups, and monitor their effectiveness. Explain your rationale for the design choices.	Creating a GPO plan to enforce security settings **Step 1:** Identify security requirements Determine the settings required for all users (e.g., password policies, software restrictions, desktop configurations) and any department-specific rules. **(3 Marks)** **Step 2:** Structure GPOs Create separate GPOs for general company-wide policies and for department-specific needs. Use starter GPOs as templates for consistency. **(3 Marks)** **Step 3:** Link GPOs to OUs or groups Apply GPOs at the OU level to target specific departments, and at the domain level for policies affecting all users. Ensure proper hierarchy to avoid conflicts. **(3 Marks)** **Step 4:** Configure and test GPO settings Use the GPO Editor to define settings. Test in a controlled environment

	QUESTIONS	Answers
		before deployment to ensure expected behaviour.**(3 Marks)** **Step 5:** Monitor and maintain Regularly review GPO effectiveness, check for conflicts, and update policies as organizational needs change. Use the Group Policy Management Console (GPMC) to monitor and manage settings efficiently. **(3 Marks)**
93.	Design and create a complete IIS Web Server configuration plan for a website that supports HTTP, HTTPS, and FTP access**.** Your design should include handler mappings**,** connection settings**,** site bindings for HTTP and HTTPS**,** and clearly explain how each configuration ensures secure and efficient web service delivery.	IIS Web Server Configuration Plan **Step 1**: Configure Handler Mappings • Identify the types of requests the server will handle (e.g., .aspx, .php, .html). • Map each file type to the appropriate handler (e.g., ASP.NET, PHP). • Ensure proper execution permissions for each handler to process requests securely. **(3 Marks)** **Step 2:** Set Connection Settings • Define maximum simultaneous connections to manage server load. • Configure connection timeout to prevent idle sessions from consuming resources.

	QUESTIONS	Answers
		• Enable logging of connection tasks for monitoring and troubleshooting.**(3 Marks)** **Step 3:** Configure FTP Access • Enable FTP protocol for file transfers. • Set up FTP user accounts with restricted permissions. • Configure FTP firewall support and SSL for secure file transfer.**(3 Marks)** **Step 4:** Configure Site Bindings • HTTP Binding: Assign IP address, port 80, and hostname (if needed). • HTTPS Binding**:** Assign IP address, port 443, and link to an SSL/TLS certificate. • Ensure correct hostname mapping if multiple sites are hosted (host headers). • Test both bindings to ensure the site is accessible via HTTP and HTTPS. **(3 Marks)** **Step 5: Verify and Test** • Browse the website via HTTP and HTTPS to confirm correct bindings.

	QUESTIONS	Answers
		• Test handler mappings by accessing different file types. • Verify FTP access using configured user credentials.**(3 Marks)**
94.	Design a complete deployment plan for a web application, considering both frontend and backend requirements. Your plan should include: • Selection of a suitable hosting platform (free or paid) based on the technical requirements. • Steps to verify the web app locally before deployment. • Configuration of backend technologies on IIS (e.g., PHP environment variables, WinCache, FastCGI for PHP, IISNode for Node.js). • Measures to ensure web app environment security. Explain and justify each configuration choice in your deployment plan.	Deployment Plan for Web Application **Step 1:** Select Hosting Platform • Analyze technical requirements: ❖ Without backend: static web app → free hosting may suffice (e.g., GitHub Pages, Netlify). ❖ With backend: dynamic web app → paid hosting recommended (e.g., IIS on Windows Server, Azure). • Choose based on scalability, security, and backend support. **(3 Marks)** **Step 2:** Verify Web App Locally • Run the web app on a local server (e.g., IIS Express, XAMPP). • Test all pages, links, forms, and API calls.

	QUESTIONS	Answers
		• Confirm backend functionality (database connections, server-side scripts). **(3 Marks)** **Step 3:** Configure Backend Technologies in IIS 1. **PHP Web App:** ❖ Set PHP environment variables. ❖ Configure **WinCache** for caching PHP scripts. ❖ Configure FastCGI handler mapping to process PHP requests. 2. Node.js Web App: ❖ Install and configure IISNode module to handle Node.js applications. **(3 Marks)** **Step 4:** Configure Environment Security • Enable HTTPS and configure SSL/TLS certificates. • Set correct folder and file permissions. • Disable directory browsing and unnecessary modules.

	QUESTIONS	Answers
		• Apply firewall and network security rules to protect the web app. **(3 Marks)** **Step 5:** Deploy Web App to Server • Copy verified local web app to the hosting server. • Configure site bindings for HTTP/HTTPS as needed. • Test all frontend and backend functionalities on the live server. **(3 Marks)** **Step 6: Monitor and Maintain** • Enable logging of requests and errors. • Monitor server performance and resource usage. • Apply security patches and updates regularly. **(3 Marks)**
95.	Design a step-by-step plan to deploy a PHP-based web application on an IIS web server	**Step 1**: Install IIS and Enable Required Features • Install IIS Role on Windows Server. • Enable HTTP Features, ASP.NET, CGI Interface, FTP Feature, and HTTP Health and Diagnostics.

	QUESTIONS	Answers
		• Confirm IIS installation and verify successful installation. **(3 Marks)** **Step 2:** Configure Backend Technology for PHP • Set PHP environment variables. • Configure FastCGI handler mapping to process PHP scripts. • Configure WinCache for caching to improve performance. **(3 Marks)** **Step 3:** Configure Site Bindings • Assign IP address, port number, and protocol (HTTP/HTTPS) to the web application. • Configure DNS records to resolve the domain name to the server IP. **(3 Marks)** **Step 4:** Upload and Test Web Application • Specify the physical path for the application files and upload the web app. • Test local URL accessibility, browser compatibility, website speed, and security settings.**(3 Marks)**

	QUESTIONS	Answers
96.	Create a strategy for hosting a web application accessible both locally and online	**Step 1:** Choose Hosting Platform • Paid hosting is preferred for production due to better security, scalability, and uptime. • Free hosting can be used for testing or small-scale projects. **(2 Marks)** **Step 2:** Configure Server Environment • Enable necessary IIS features (HTTP, ASP.NET, CGI, FTP). • Configure backend technologies depending on the application type (PHP, Node.js, ASP.NET). **(2 Marks)** **Step 3:** Upload Application and Configure DNS • Upload web application to the server. • Specify physical path, select protocol, assign IP address, configure port number. • Set DNS records to allow online access. **(3 Marks)**

	QUESTIONS	Answers
		Step 4: Testing and Verification • Test local network accessibility. • Test online accessibility from multiple devices and browsers. • Check website speed, file visibility, and security. **(3 Marks)** **Outcome:** The web application is available both locally and online with secure and efficient configuration.
97.	Develop a plan to manage multiple web applications on a single IIS server	**Step 1:** Install and Verify IIS Roles • Install IIS Role and enable HTTP, ASP.NET, CGI, FTP, and diagnostics features. • Confirm IIS installation for server readiness. **(3 Marks)** **Step 2:** Configure Separate Site Bindings • Assign unique IP addresses, ports, and protocols (HTTP/HTTPS) for each web application. • Configure DNS entries for each domain or subdomain.

	QUESTIONS	Answers
		(3 Marks) **Step 3**: Configure Backend Technologies • Configure PHP (FastCGI, WinCache) for PHP apps. • Enable IISNode module for Node.js applications. • Configure ASP.NET settings for ASP.NET applications. **(3 Marks)** **Step 4**: Monitoring and Troubleshooting • Use IIS Manager to monitor handler mappings, connection tasks, and FTP operations. • Verify local and online accessibility, website speed, and security regularly. • Handle discovered errors promptly to ensure uptime and performance. **(3 Marks)** Outcome: Efficient management of multiple applications on a single server with proper backend support, security, and scalability.

	QUESTIONS	Answers
98.	Design a plan to deploy a PHP web application on a Windows Server using IIS. Include steps for installing required IIS features, configuring backend technology, setting site bindings, and ensuring proper security and accessibility.	Deploying a PHP web application on Windows Server using IIS **Step 1:** Install IIS and required features • Enable IIS role via Server Manager. • Enable HTTP Features**,** ASP.NET**,** CGI**,** and FTP **i**f needed. **(3 Marks)** **Step 2:** Configure backend technology • Install PHP and configure PHP environment variables. • Set up FastCGI handler mapping for PHP. • Configure caching if required (e.g., WinCache). **(3 Marks)** **Step 3:** Set up site bindings • Assign the website to a specific IP address, port number, and protocol (HTTP or HTTPS). • Configure SSL certificates if HTTPS is used. **(3 Marks)** **Step 4:** Ensure security and accessibility • Apply necessary file and folder permissions.

	QUESTIONS	Answers
		• Configure firewall rules to allow traffic. • Test website accessibility locally and online. **(3 Marks)** **Step 5:** Verify deployment Confirm that the web app runs correctly, check speed, browser compatibility, and website size. **(3 Marks)**
99.	Create a strategy to host a web application for a medium-sized organization that ensures high availability, fast load times, and secure access. Include considerations for server environment, hosting platform (free vs paid), protocols, DNS, and testing.	Hosting a web application with high availability and secure access **Step 1:** Choose hosting platform • Decide between paid IIS hosting for production-critical apps or free hosting for testing/small apps. **(3 Marks)** **Step 2:** Configure server environment • Set up IIS with necessary features (HTTP, ASP.NET, CGI, FTP). • Install backend technologies (PHP, Node.js) as required. **(3 Marks)** **Step 3:** Assign protocols, IPs, and DNS

	QUESTIONS	Answers
		• Configure site bindings for HTTP/HTTPS. • Assign IP addresses and configure DNS settings. • Choose appropriate port numbers for each application. **(3 Marks)** **Step 4:** Implement security measures • Set folder permissions, enable HTTPS, and configure firewall rules. • Monitor server and application for vulnerabilities. **(3 Marks)** **Step 5:** Test and optimize • Test accessibility locally and online. • Check website speed, browser compatibility, and file size. • Optimize performance (caching, resource allocation). **(3 Marks)**
100.	Design a Group Policy and IIS configuration plan to manage multiple web applications on a single Windows Server. Include how you would separate applications, assign protocols and ports, handle DNS, and maintain security for each application.	Managing multiple web applications on a single Windows Server **Step 1:** Separate applications • Create individual sites in IIS **f**or each web application.

	QUESTIONS	Answers
		• Assign unique physical paths for each site. **(3 Marks)** **Step 2:** Assign protocols and ports • Configure site bindings with distinct IP addresses, ports, and protocols (HTTP/HTTPS) for each application. **(3 Marks)** **Step 3:** Configure DNS • Map each web application to a unique domain or subdomain. • Ensure proper resolution for internal and external access. **(3 Marks)** **Step 4:** Apply security measures • Set file/folder permissions specific to each application. • Enable HTTPS and firewall rules to protect applications. **(3 Marks)** **Step 5:** Monitor and maintain • Test accessibility, website speed, and functionality for each application.

	QUESTIONS	Answers
		• Use logs and monitoring tools to manage performance and troubleshoot issues. **(3 Marks)**